

Pharmacovigilance ICSR Processing Portfolio

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Introduction:

This project presents a set of processed literature ICSRs developed to demonstrate practical understanding of pharmacovigilance case processing and safety data handling.

Purpose:

This portfolio demonstrates structured, guideline-aligned Pharmacovigilance case processing including –

ICSR Case Processing Workflow:

1. Literature search & Case selection
2. ICSR Intake & Validation
3. MedDRA Coding
4. Seriousness Assessment
5. Expectedness Assessment
6. Causality Assessment (WHO-UMC, Naranjo Scale)
7. Narrative Writing

Methodology:

Four published literature-based Individual Case Safety Reports (ICSRs) were selected from PubMed. Relevant clinical information was extracted and systematically processed, including patient details, suspect drug, adverse event, MedDRA coding, seriousness, expectedness, WHO-UMC causality assessment, and regulatory narrative writing, following standard pharmacovigilance practices.

Literature-Based ICSR Case Processing:

- Case 01: Amoxicillin Induced Fever, Rash, and Catatonia
- Case 02: Atorvastatin-Induced Rhabdomyolysis
- Case 03: Metformin-Associated Lactic Acidosis
- Case 04: ACE Inhibitor-Induced Cough

Key Learnings:

- Learned end-to-end ICSR case processing.
- Applied MedDRA coding and WHO-UMC causality assessment.
- Developed regulatory narrative writing skills.
- Improved understanding of pharmacovigilance practices.

Conclusion:

This portfolio demonstrates my practical knowledge of literature-based ICSR case processing and strengthens my understanding of pharmacovigilance principles and drug safety reporting.

CASE 01: Amoxicillin Induced Fever, Rash, and Catatonia - A Case Study

Source Information

- PMID: 36892033
- Source Type: Literature ICSR

Patient Information

- Age: 23
- Gender: Female
- Medical history: Postpartum

Drug Information

- Suspect Drug: Amoxicillin
- Dose: Amoxicillin-Clavulanic Acid (Amoxiclav) 625 mg
- Route: Injection + Oral tablets
- Indication: episiotomy wound infection following delivery.
- Frequency: Not reported
- Start/Stop date: Not reported
- WHO-DD / ATC Code: J01CR02

Adverse Events

- Altered sensorium, Fever followed by maculopapular rash, Catatonia and Vasculitic rash
- Onset: After initiation of Amoxiclav therapy
- Treatment for AE: Lorazepam challenge (positive response)
- Outcome: Improved (recovered)

Validity Assessment

A valid ICSR requires:

1. Identifiable Patient: 23-year-old Female
2. Identifiable Reporter: Literature Author (HCP)
3. Suspect Drug: Amoxicillin
4. Adverse Event: Altered sensorium, Fever followed by maculopapular rash, generalized rigidity, Catatonia and Vasculitic rash

MedDRA Coding

Lowest Level Term (LLT)	Preferred Term (PT)	High Level Term (HLT)	High Level Group Term (HLGT)	System Organ Class (SOC)
Altered sensorium	Altered state of consciousness	Disturbance in consciousness NEC	Neurological disorders NEC	Nervous system disorders

Fever	Pyrexia	Body temperature conditions and changes NEC	Body temperature conditions	General Disorders and Administration Site Conditions
Rash maculopapular	Rash maculopapular	Rashes, eruptions and exanthems NEC	Epidermal and dermal conditions	Skin and Subcutaneous Tissue Disorders
Generalized rigidity	Muscle rigidity	Movement disorders NEC	Movement disorders (incl parkinsonism)	Nervous System Disorders
Catatonia	Catatonia	Schizophrenia and related disorders	Psychotic disorders	Psychiatric Disorders
Vasculitic rash	Vasculitic rash	Vascular disorders NEC	Vascular disorders	Skin and Subcutaneous Tissue Disorders

Seriousness Assessment

- Seriousness Status: Serious
- Seriousness Criteria Met: Hospitalization; Important Medical Event (IME)
- Rationale: Catatonia is considered a medically significant neurological event and may require urgent medical intervention.

Expectedness Assessment

- For fever followed by maculopapular rash: Expected
- For Catatonia and vasculitic rash: Unexpected Rare adverse effect; not commonly expected

Causality Assessment (WHO-UMC)

- Probable: Temporal relationship with Amoxiclav; improved with lorazepam; no alternative cause identified

Narrative Writing

A literature report described a 23-year-old postpartum female who received empirical amoxicillin-clavulanic acid (Amoxiclav) 625 mg by injection followed by oral tablets for the management of an episiotomy wound. The patient had no relevant medical history reported in the available literature. Following initiation of amoxicillin-clavulanic acid therapy, the patient developed fever and altered sensorium, which was subsequently followed by a maculopapular rash. On clinical examination, generalized rigidity with waxy flexibility was observed. A lorazepam challenge test resulted in clinical improvement, confirming the diagnosis of catatonia. Further evaluation identified amoxicillin-clavulanic acid as the suspected precipitating factor for the adverse events.

The suspect drug was discontinued, and the patient received appropriate medical management. The reported adverse events included drug-induced catatonia, fever, and maculopapular rash. The clinical condition improved following treatment and withdrawal of the suspect drug (positive dechallenge). Rechallenge information was not reported.

Based on the temporal association between drug administration and symptom onset, positive dechallenge, and the absence of an alternative confirmed cause in the available report, the causal relationship was assessed as Probable according to the WHO-UMC causality assessment. This serious literature Individual Case Safety Report (ICSR) was identified from a published case report describing a rare adverse reaction associated with amoxicillin-clavulanic acid in a postpartum patient.

Medical Review

- Temporal Association: Positive – Symptoms developed after initiation of amoxicillin.
- Dechallenge: Positive – Symptoms improved after discontinuation of amoxicillin.
- Rechallenge: Not Reported
- Alternative Cause: Not Established
- Dose-Response: Not applicable

Quality Check

- Validity confirmed: Yes
- Seriousness assessed: Yes
- MedDRA coding complete: Yes
- Narrative completed: Yes
- Causality documented: Yes
- Source referenced: Yes
- Listedness assessed: Yes — Unlisted (catatonia & vasculitic rash); confirmed against product label
- Submission timeline noted: Yes — Serious + Unexpected → 15-day expedited report; Day 0 = date of receipt

Key Findings

- Suspect drug: Amoxicillin
- Adverse events: Altered sensorium, Fever followed by maculopapular rash, generalized rigidity, Catatonia and Vasculitic rash
- Case type: Literature
- Seriousness: Serious
- Expectedness: Mixed
- Causality: Probable
- Special population: Postpartum female
- Outcome: Improved (recovered)

Signal Relevance

Catatonia is an unusual neurological event and would warrant further review for signal detection purposes if additional similar reports were identified.

CASE 02: High-Intensity Atorvastatin-Induced Rhabdomyolysis in an Elderly Patient With NSTEMI: A Case Report and Review of the Literature

Source Information

- PMID: 27798246
- Source Type: Literature ICSR

Patient Information

- Age: 91
- Gender: Male
- Medical history: Myocardial infarction (4 months prior to admission)

Drug Information

- Suspect Drug: Atorvastatin
- Dose: 80 mg daily (initiated post-MI); reduced to 40 mg daily as instructed, but non-adherent until 10 days prior to admission
- Route: Oral
- Indication: Post-MI secondary prevention (cholesterol management)
- Frequency: Once daily
- Start date: 4 months prior to admission
- WHO-DD/ATC Code: C10AA05

Adverse Events

- Muscle weakness, muscle pain, unexplained soreness, elevated CPK (18,723 U/L), elevated serum creatinine (1.6 mg/dL)
- Onset: 10 days prior to admission (coinciding with continued high-dose use beyond instructed reduction)
- Treatment for AE: Atorvastatin held; IV fluids administered
- Outcome: Not explicitly stated — improving with supportive treatment

Validity Assessment

A valid ICSR requires:

1. Identifiable Patient: 91-year-old Male
2. Identifiable Reporter: Literature Author (HCP)
3. Suspect Drug: Atorvastatin
4. Adverse Event: Muscle weakness, muscle pain, rhabdomyolysis, elevated CPK, acute kidney involvement (elevated creatinine)

MedDRA Coding

LLT	PT	HLT	HLGT	SOC
Muscle weakness	Muscular weakness	Muscle disorders NEC	Muscle disorders	Musculoskeletal and connective tissue disorders
Muscle pain	Myalgia	Muscle disorders NEC	Muscle disorders	Musculoskeletal and connective tissue disorders
Rhabdomyolysis	Rhabdomyolysis	Muscle disorders NEC	Muscle disorders	Musculoskeletal and connective tissue disorders
Elevated CPK	Blood creatine phosphokinase increased	Enzyme investigations NEC	Cardiac and vascular investigations	Investigations
Elevated serum creatinine	Blood creatinine increased	Renal function analyses	Renal and urinary investigations and procedures	Investigations

Seriousness Assessment

- Seriousness Status: Serious
- Seriousness Criteria Met: Hospitalization (explicitly confirmed — patient was admitted); Medically Significant Event
- Rationale: Rhabdomyolysis with markedly elevated CPK (18,723 U/L) and renal involvement (creatinine 1.6 mg/dL) required hospital admission and IV fluid therapy — meets clear hospitalization criterion.

Expectedness Assessment

- Rhabdomyolysis/myopathy: Expected — well-documented, labeled ADR for statins, especially at high doses.
- Severity at this dose/age combination: This case illustrates dose- and age-related risk amplification rather than a novel reaction.

Causality Assessment (WHO-UMC)

- Probable: Strong temporal relationship (symptom onset 10 days after resuming non-adherent high-dose therapy); positive dechallenge (improvement after holding drug); known mechanism (statin-induced myotoxicity); no alternative cause identified.

Narrative Writing

A literature report described a 91-year-old male patient with a medical history of myocardial infarction who was initiated on atorvastatin 80 mg orally once daily for secondary prevention following the cardiac event. Although the treating physician advised reducing the atorvastatin dose to 40 mg daily three months prior to admission, the patient did not adhere to the dose adjustment until 10 days before hospitalization. No concomitant medications were reported. The patient presented with worsening muscle weakness, muscle pain, and unexplained soreness of 10 days' duration. On hospital admission, laboratory investigations showed a markedly elevated serum creatinine phosphokinase level of 18,723 U/L and a serum creatinine level of 1.6 mg/dL, leading to the diagnosis of rhabdomyolysis. Atorvastatin was discontinued (positive dechallenge), and the patient received supportive treatment with intravenous fluids. The clinical outcome following treatment was reported as improvement; however, the final recovery status was not specified. Rechallenge information was not reported.

Based on the temporal relationship between atorvastatin exposure and the onset of rhabdomyolysis, along with improvement following discontinuation of the suspect drug, the causal relationship was assessed as Probable according to the WHO-UMC causality assessment. This serious literature Individual Case Safety Report (ICSR) was identified from a published case report (PMID: 27798246), reported by healthcare professionals, highlighting the increased risk of rhabdomyolysis associated with high-intensity atorvastatin therapy in elderly patients.

Medical Review

- Temporal Association: Positive — clear link between dose non-adherence and symptom onset
- Dechallenge: Positive — drug held, patient treated supportively
- Rechallenge: Not reported
- Alternative Cause: Not established
- Dose-Response Relationship: Strongly supported — patient tolerated 40 mg without issue per provider instruction; symptoms emerged specifically when reverting to 80 mg

Quality Check

- Validity confirmed: Yes
- Seriousness assessed: Yes
- MedDRA coding complete: Yes
- Narrative completed: Yes
- Causality documented: Yes
- Listedness assessed: Yes — Listed (rhabdomyolysis is a known statin ADR); confirmed against atorvastatin product label
- Source referenced: Yes

- Submission timeline noted: Yes — Serious + Listed → standard expedited/periodic reporting per regional requirement; Day 0 = date of receipt

Key Findings

- Suspect drug: Atorvastatin
- Adverse events: Muscle weakness, myalgia, rhabdomyolysis, elevated CPK, elevated creatinine
- Case type: Literature
- Seriousness: Serious
- Expectedness: Expected (Listed)
- Causality: Probable
- Special population: Geriatric (age >75) — dose-related risk factor
- Outcome: Not explicitly stated — improving with supportive treatment

Signal Relevance

This case reinforces an already well-established signal (statin-induced rhabdomyolysis, age and dose as risk modifiers) rather than identifying a new one. Its value lies in demonstrating risk-factor-driven causality reasoning — useful for periodic benefit-risk reviews and supports existing labeling/guideline recommendations regarding statin intensity in elderly populations.

CASE 03: Metformin-Associated Lactic Acidosis

Source Information

- PMID: 34244196
- Source Type: Literature ICSR

Patient Information

- Age: 58
- Gender: Female
- Medical history: Type 2 Diabetes Mellitus
- Relevant Clinical Context at Time of Event: Active diarrhoea and vomiting

Drug Information

- Suspect Drug: Metformin
- Dose: Not Reported
- Route: Oral
- Indication: Type 2 Diabetes Mellitus
- Frequency: Not Reported
- Start/Stop Date: Not Reported
- WHO-DD/ATC Code: A10BA02
- Concomitant drugs: ACE Inhibitor, NSAID
- Indication: Hypertension, Pain/inflammation

Adverse Events

- Lactic acidosis, cardiovascular instability, acute kidney injury, cardiac arrest
- Onset: Not reported — presentation at ED temporally associated with continued metformin use during acute GI illness
- Treatment for AE: Metformin discontinued; supportive care; renal replacement therapy (dialysis)
- Outcome: Improving/Recovered — rapid improvement in acid-base and haemodynamic parameters; full recovery not explicitly confirmed

Validity Assessment

A valid ICSR requires:

1. Identifiable Patient: 58-year-old Female
2. Identifiable Reporter: Literature Authors (HCP)
3. Suspect Drug: Metformin
4. Adverse Event: Lactic acidosis, cardiovascular instability, acute kidney injury, cardiac arrest

MedDRA Coding

LLT	PT	HLT	HLGT	SOC
Lactic acidosis	Lactic acidosis	Acidosis and alkalosis	Acid-base disorders	Metabolism and nutrition disorders
Acute kidney injury	Acute kidney injury	Renal failure and impairment	Renal disorders	Renal and urinary disorders
Cardiovascular instability	Haemodynamic instability	Vascular disorders NEC	Vascular disorders	Vascular disorders
Cardiac arrest	Cardiac arrest	Cardiac arrhythmias NEC	Cardiac arrhythmias	Cardiac disorders

Seriousness Assessment

- Seriousness Status: Serious
- Seriousness Criteria Met: Hospitalization, Life-threatening & Important Medical Event
- Rationale: Patient experienced a brief asystolic cardiac arrest, profound lactic acidosis (pH 6.6, lactate 14 mmol/L — critically abnormal values), cardiovascular instability, and acute kidney injury requiring renal replacement therapy (dialysis).

Expectedness Assessment

- Lactic acidosis: Expected — listed ADR for metformin, though rare; black box warning exists in prescribing information
- Cardiac arrest/asystole: Unexpected as a direct individual outcome — consequence of the severity of lactic acidosis rather than a separately listed ADR
- Acute kidney injury: Expected in context — known risk with ACE inhibitor + NSAID combination, especially during dehydration
- Overall expectedness: Mixed — primary event (lactic acidosis) is listed; severity and cardiac arrest consequence are unexpected at this extreme level

Causality Assessment (Naranjo Scale)

Question	Score	Reason
Previous reports of this reaction?	+1	Yes — well documented
AE appeared after suspect drug?	+2	Yes — AE appeared after drug
AE improved on dechallenge?	+1	Yes — improved after stopping
AE reappeared on rechallenge?	0	DNK — rechallenge not done
Alternative causes present?	-1	Yes — alternative causes exist
Placebo rechallenge?	0	DNK — no placebo

Drug detected in toxic levels?	0	DNK — plasma levels not reported
Dose-severity relationship?	0	DNK — no dose change
Similar reaction previously?	0	DNK — not reported
AE confirmed by objective evidence?	+1	Yes — pH 6.6, lactate 14, AKI, cardiac arrest
Final Score	4	Possible

Narrative Writing

A literature report described a 58-year-old female patient with a medical history of type 2 diabetes mellitus who was receiving metformin therapy along with an ACE inhibitor and a non-steroidal anti-inflammatory drug. While experiencing diarrhoea and vomiting, the patient continued taking her regular medications. She subsequently presented to the emergency department with profound lactic acidosis, cardiovascular instability, and acute kidney injury. Laboratory investigations revealed a blood pH of 6.6 and a lactate level of 14 mmol/L. The patient experienced a brief asystolic cardiac arrest and required supportive treatment, including renal replacement therapy. Metformin was discontinued, and the patient's acid-base abnormalities and haemodynamic parameters improved rapidly following treatment. Based on the temporal association, positive dechallenge, and the known association between metformin and lactic acidosis, the causal relationship was assessed as Possible according to the causality assessment (Naranjo's Scale). This literature report represents a serious Individual Case Safety Report (ICSR) was identified from a published case report (PMID: 34244196), reported by healthcare professionals, due to its life-threatening nature and hospitalization.

Medical Review

- Temporal Association: Positive — metformin continued during acute GI illness; dehydration impaired renal clearance, causing metformin accumulation
- Dechallenge: Positive — improvement with metformin cessation + RRT
- Rechallenge: Not reported
- Alternative Cause: Multi-factorial — ACE inhibitor and NSAID concomitant use + dehydration from GI illness contributed to acute kidney injury, which in turn reduced metformin clearance. Metformin remains primary suspect.
- Dose-Response: Not applicable — standard therapeutic dose; accumulation due to impaired elimination, not dose escalation

Quality Check

- Validity confirmed: Yes
- Seriousness assessed: Yes — three criteria met (hospitalization, life-threatening, IME)
- MedDRA coding complete: Yes
- Narrative completed: Yes
- Causality documented: Yes — Naranjo scale (Possible/4)

- Listedness assessed: Yes — lactic acidosis is listed, cardiac arrest severity noted as exceeding typical labeling description
- Source referenced: Yes
- Concomitant medications documented: Yes — ACE inhibitor + NSAID noted
- Submission timeline noted: Yes — Serious + Listed → expedited per company SOP; Day 0 = date of receipt

Key Findings

- Suspect drug: Metformin
- Concomitant drugs: ACE inhibitor + NSAID
- Adverse events: Adverse events: Lactic acidosis, cardiovascular instability, acute kidney injury, asystolic cardiac arrest
- Case type: Literature
- Seriousness: Serious
- Expectedness: Mixed (listed primary event; unexpected severity/cardiac arrest)
- Causality: Possible (Naranjo score 4)
- Special feature: Drug-drug-disease interaction
- Outcome: Improving/Recovered

Signal Relevance

This case supports the known link between metformin and lactic acidosis, especially when a patient is dehydrated from illness or taking drugs that affect the kidneys. It highlights a key risk-management priority: patient education on "sick-day rules" (stopping metformin during periods of vomiting, diarrhea, or dehydration to prevent drug accumulation).

CASE 04: ACE Inhibitor-Induced Cough

Source Information

- PMID: 2542908
- Source Type: Literature (ICSR)

Patient Information

- Age: 65
- Gender: Female
- Medical history: Hypertension; Preexisting cystocele; Congestive heart failure
- Special Population: Elderly female (age and sex noted as potential contributing factors per authors)

Drug Information

Field	Suspect Drug 1	Suspect Drug 2
Drug Name	Enalapril	Captopril
Role	Primary suspect	Secondary suspect (rechallenge with related drug)
Dose	2.5 mg daily	6.25 mg twice daily
Route	Oral	Oral
Indication	Hypertension	Congestive heart failure
Frequency	Once daily	Twice daily
Start date	Not reported	During subsequent hospitalization
Stop date	12 days after initiation (approx)	Within 24 hours of initiation
WHO-DD/ATC Code	C09AA02	C09AA01

Adverse Events

- Persistent dry cough, stress incontinence (secondary to cough), functional decline

Field	Suspect Drug 1(Enalapril)	Suspect Drug 2(Captopril)
Onset	12 days after initiation of drug	12 days after initiation of drug
Treatment of AE	Discontinuation of the drug	Discontinuation of the drug
Outcome	Both the coughing and the stress incontinence resolved completely after the medication was stopped	The cough resolved completely after the medication was stopped

Validity Assessment

A valid ICSR requires:

1. Identifiable Patient: 65-year-old Female
2. Identifiable Reporter: Literature Authors (HCP)
3. Suspect Drug: Enalapril + Captopril (ACE inhibitors)
4. Adverse Event: Persistent dry cough, stress incontinence, functional decline

MedDRA Coding

LLT	PT	HLT	HLGT	SOC
Dry cough	Cough	Coughing and associated symptoms	Respiratory and mediastinal disorders NEC	Respiratory, thoracic and mediastinal disorders
Stress incontinence	Stress urinary incontinence	Bladder and urethral symptoms	Urinary tract signs and symptoms	Renal and urinary disorders
Functional decline	Functional ability decreased	Activities of daily living NEC	Lifestyle issues	General disorders and administration site conditions

Seriousness Assessment

- Seriousness Status: Non-serious
- Seriousness Criteria Met: None of the standard ICH E2A criteria met, the AE did not result in death, hospitalization, disability, or other seriousness criteria.
- Rationale: Dry cough and secondary stress incontinence, while significantly impacting quality of life, do not meet formal seriousness criteria. The subsequent hospitalization was for congestive heart failure — not for the cough itself.

Expectedness Assessment

- Dry cough: Listed/Expected— ACE inhibitor-induced cough is a well-recognized, labeled class-effect ADR
- Stress incontinence: Unlisted/Unexpected — secondary consequence of cough in context of preexisting cystocele; not a directly labeled ADR for ACE inhibitors
- Overall expectedness: Mixed—primary AE listed; secondary AE unlisted

Causality Assessment (Naranjo Scale)

Question	Score	Reason
Previous reports exist?	+1	Yes — ACE inhibitor cough well documented in literature
AE appeared after drug?	+2	Yes — Cough appeared 12 days after enalapril
AE improved on dechallenge?	+1	Yes — Resolved on enalapril discontinuation
AE reappeared on rechallenge?	+2	Yes — Recurred within 24hrs of captopril
Alternative causes present?	+2	No — No alternative cause identified
Placebo rechallenge?	0	DNK — Not applicable
Drug in toxic concentration?	0	DNK — Not reported
Dose-severity relationship?	0	DNK — Not reported
Prior similar reaction?	0	DNK — Not reported
Confirmed by objective evidence?	0	No — Cough is a clinical symptom — no lab confirmation
Total	8	Probable

Narrative Writing

A literature report described a 65-year-old female patient with a medical history of hypertension, cystocele, and congestive heart failure who was initiated on enalapril 2.5 mg orally once daily for hypertension. Twelve days after starting therapy, the patient developed a persistent dry cough, which subsequently led to stress urinary incontinence due to her pre-existing cystocele and resulted in a decline in functional status. Enalapril was discontinued, following which both the cough and urinary incontinence resolved (positive dechallenge). During a subsequent hospitalization, the patient was prescribed captopril 6.25 mg orally twice daily for congestive heart failure. Within 24 hours of initiating captopril, the dry cough recurred (positive rechallenge) and resolved again after discontinuation of the drug. Based on the positive temporal association, positive dechallenge, positive rechallenge, and absence of an alternative cause, the causal relationship was assessed as Certain according to the Naranjo Scale causality assessment. This literature report represents a valid non-serious Individual Case Safety Report (ICSR) (PMID: 2542908) demonstrating a class-related adverse reaction associated with angiotensin-converting enzyme inhibitors.

Medical Review

- Temporal Association: Positive — cough onset 12 days after enalapril
- Dechallenge: Positive — resolved on stopping enalapril
- Rechallenge: Positive — recurred within 24 hours of captopril
- Alternative Cause: Not established
- Class Effect: Confirmed — reaction occurred with both enalapril and captopril, supporting pharmacological class-effect mechanism

Quality Check

- Validity confirmed: Yes
- Seriousness assessed: Yes — Non-serious; criteria clearly not met
- MedDRA coding complete: Yes
- Narrative completed: Yes
- Causality documented: Yes — Naranjo scale (Probable/8)
- Listedness assessed: Yes — cough listed; stress incontinence unlisted
- Source referenced: Yes
- Submission timeline noted: Yes — Non-serious + Listed → periodic reporting (PSUR/PBRER)

Key Findings

- Suspect drugs: Enalapril + Captopril (ACE inhibitor class)
- Adverse events: Dry cough, stress urinary incontinence & functional decline
- Case type: Literature
- Seriousness: Non-serious
- Expectedness: Mixed (cough listed; incontinence unlisted)
- Causality: Probable/8 (Naranjo)
- Special feature: Positive rechallenge
- Outcome: Resolved / Recovered

Signal Relevance

This case contributes to the well-established signal of ACE inhibitor-induced cough as a pharmacological class effect. Its unique value lies in demonstrating the class-effect mechanism — the reaction occurring with both enalapril and captopril confirms it is attributable to ACE inhibition itself rather than to drug-specific properties. Additionally, this case illustrates how a seemingly minor ADR (cough) can cascade into secondary complications (stress incontinence, functional decline) with significant quality-of-life impact — a principle directly relevant to benefit-risk assessment and patient-centered pharmacovigilance.